

AI-Based Phishing URL Detection

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Introduction

Phishing attacks are a major cybersecurity threat that aim to steal sensitive user information through malicious URLs. Traditional detection methods struggle to identify new attacks, creating a need for fast and intelligent solutions. This project uses machine learning and URL-based features to enable accurate and real-time phishing detection.

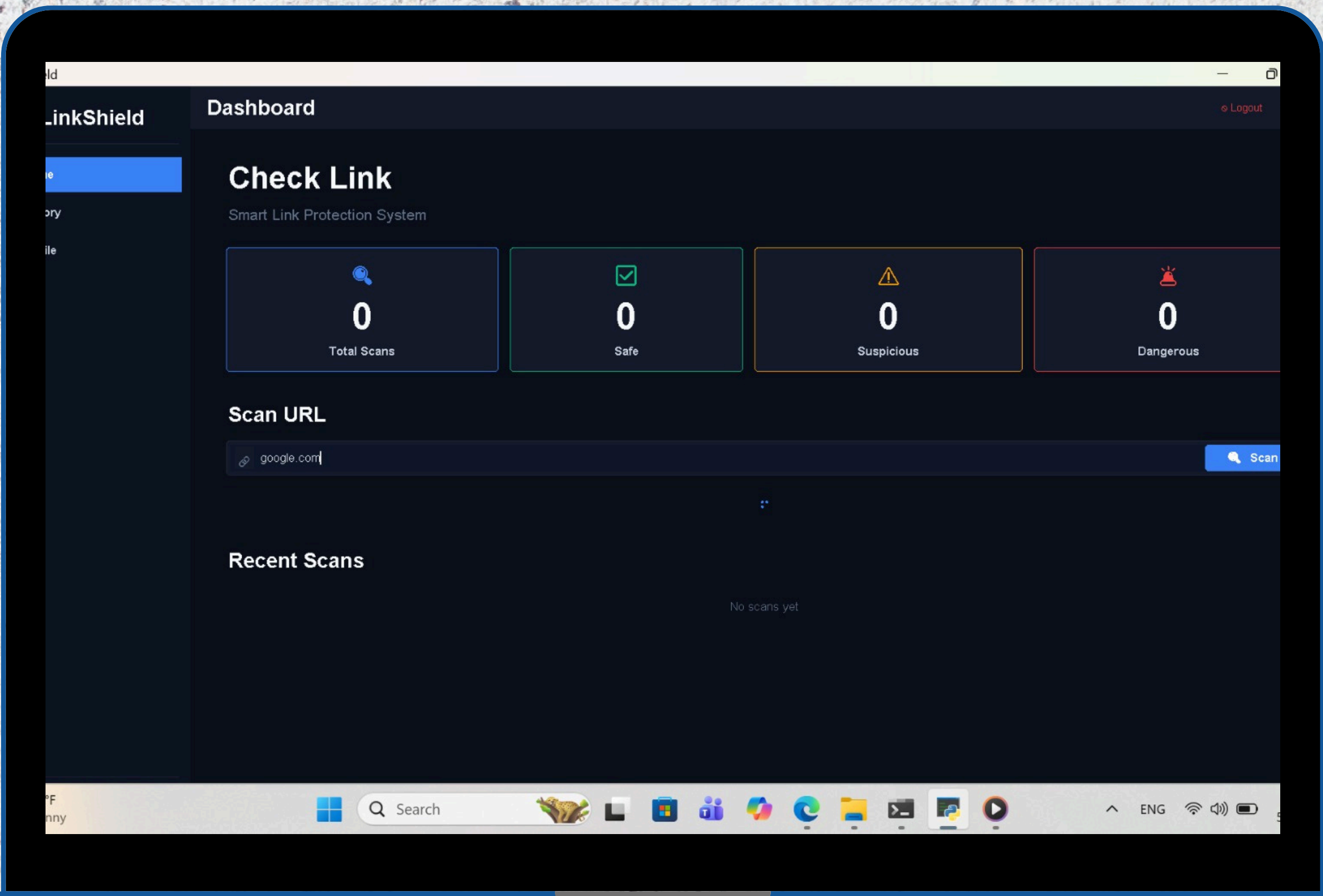
problem Statement

- Phishing attacks evolve rapidly and bypass traditional methods
- Blacklist-based systems cannot detect new (zero-day) attacks
- Existing solutions are either slow or resource-intensive
- Need for a fast, accurate, and real-time detection system

Methodology

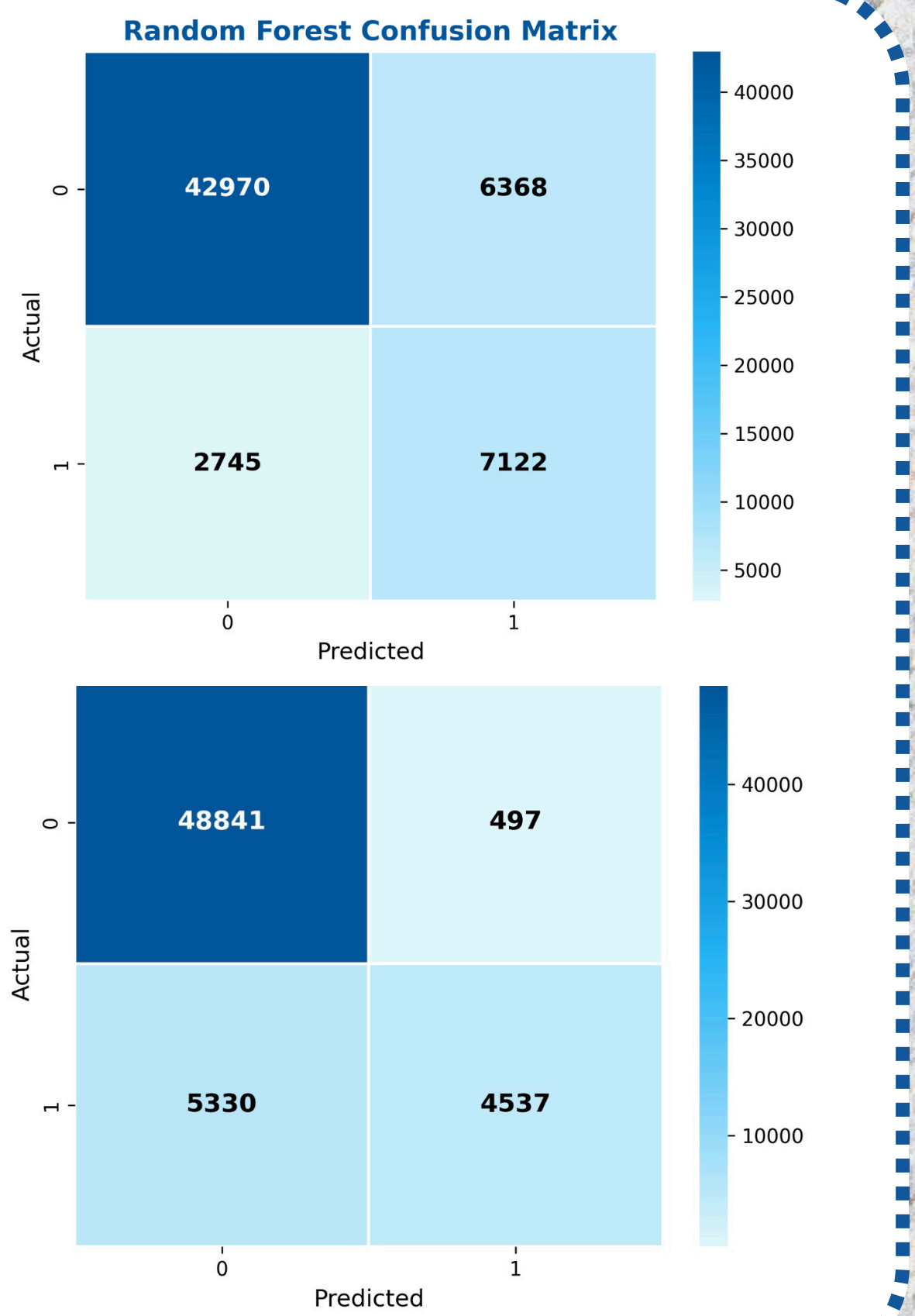
- Dataset:
- PhishTank (phishing URLs)
- Tranco Top 1M (legitimate URLs)
- Data preprocessing and balancing
- Feature extraction (7 lexical features)
- Models:
- Random Forest
- Gradient Boosting

Interface



Results

- Dataset:
- PhishTank & Tranco Top 1M
- Data split:
- 80% Training / 20% Testing
- Models:
- Random Forest (RF)
- Gradient Boosting (GB)
- Results:
- RF Recall: **72%** (better phishing detection)
- GB Precision: **90%** (fewer false positives)
- Accuracy:
- RF: **85%**
- GB: **90%**
- Inference Time: **< 0.2 ms per URL**



Conclusion

The proposed system demonstrates that combining lexical feature engineering with lightweight machine learning can effectively detect phishing URLs with high speed and acceptable accuracy, making it suitable for real-world applications.

After entering a URL, click the “Scan” button to display the result.

